

SDG12: Ensure sustainable consumption and production patterns

SUMMARY

Original search string: 1,367,789 document results

Original Query

TITLE-ABS-KEY(((environmental justice} OR {environmental damage} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR pollution OR pollutant* OR ozone OR pesticide* OR {life cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {natural resource} OR {natural resources} OR {resource footprint} OR {ecological footprint} OR {solid waste} OR {transportation cost} OR {transportation costs} OR {transport cost} OR {transport costs} OR post-harvest OR {waste generation} OR {corporate report} OR {corporate reports} OR {corporate sustainability} OR consumption OR production OR {business practice} OR {business practices} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR deforest* OR recycl* OR touris* OR {natural product} OR {natural products} OR {biobased product} OR {biobased products} OR {bio-based product} OR {bio-based products} OR {supply chain} OR {zero waste})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Graphical summary of co-occurrence of terms from 2000 randomly selected results:

A graphic summary was not possible as the query returned that many results that no random sample could be drawn by using the SDG analytics tool (time out).

Third and final update search: 68,509 document results

TITLE-ABS-KEY ({environmental pollution} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR {chemical pollution} OR {ozone depletion} OR {pesticide pollution} OR {pesticide stress} OR {pesticide reduction} OR {life cycle assessment} OR {life cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {environmental footprint} OR {material footprint} OR {harvest efficiency} OR {solid waste} OR {waste generation} OR {corporate social responsibility} OR {corporate sustainability} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR {waste recycling} OR {resource recycling} OR {resource reuse} OR {biobased economy} OR {zero waste} OR {sustainability label} OR {sustainability labelling} OR {global resource extraction} OR {material flow accounting} OR {societal metabolism} OR {food spill} OR {resource spill} OR {resource efficiency} OR {sustainable food consumption} OR {green consumption} OR {sustainable supply chain} OR {circular economy} OR {cradle to cradle} OR {sustainable procurement} OR {sustainable tourism} OR {fossil-fuel subsidies} OR {fossil-fuel expenditure} OR {sustainability label} OR {sustainability labelling} OR (consumption AND ({resource use} OR spill)) OR (production AND ({resource use} OR spill)) AND NOT ({wireless sensor network} OR {wireless sensor networks} OR {wireless network} OR {wireless networks} OR {wireless} OR {disease} OR {astrophysics})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

PROCESS

The first stage of this review was to compare the original search string to the targets listed under the Sustainable Development Goal (SDG).

SDG targets listed here: <https://sustainabledevelopment.un.org/sdg12>

The purpose of this stage was to understand whether the Scopus results correctly reflected the theme of the SDG. The SDGs were unanimously agreed by member states of the United Nations, with very

deliberate language and scope/ exclusions, so the results of this analysis should accurately reflect the specific SDG themes.

The theme of SDG 12 promotes **sustainable resource use** and **waste reduction**. A sustainable use of natural resources by **limiting food – and resources losses** should be promoted. Health - and environmental impacts should be limited by **reducing the release of chemicals and waste into the environment** (air, water, soil) at each stage of a supply chain, from producer to consumer. **Corporate social responsibility** is promoted. **Recycling** and reuse of resources should be promoted.

A Scopus search was conducted using the original search string. This returned 1,367,789 document results.

The top 20 cited articles were reviewed. 18 of the 20 articles do not closely reflect the theme of the SDG. Specifically, these related to energy sciences, medicine, IT, astrophysics, deforestation.

Top 20 most cited articles listed in Scopus from the original search string:

*red text shows articles which do not closely match the theme of the SDG

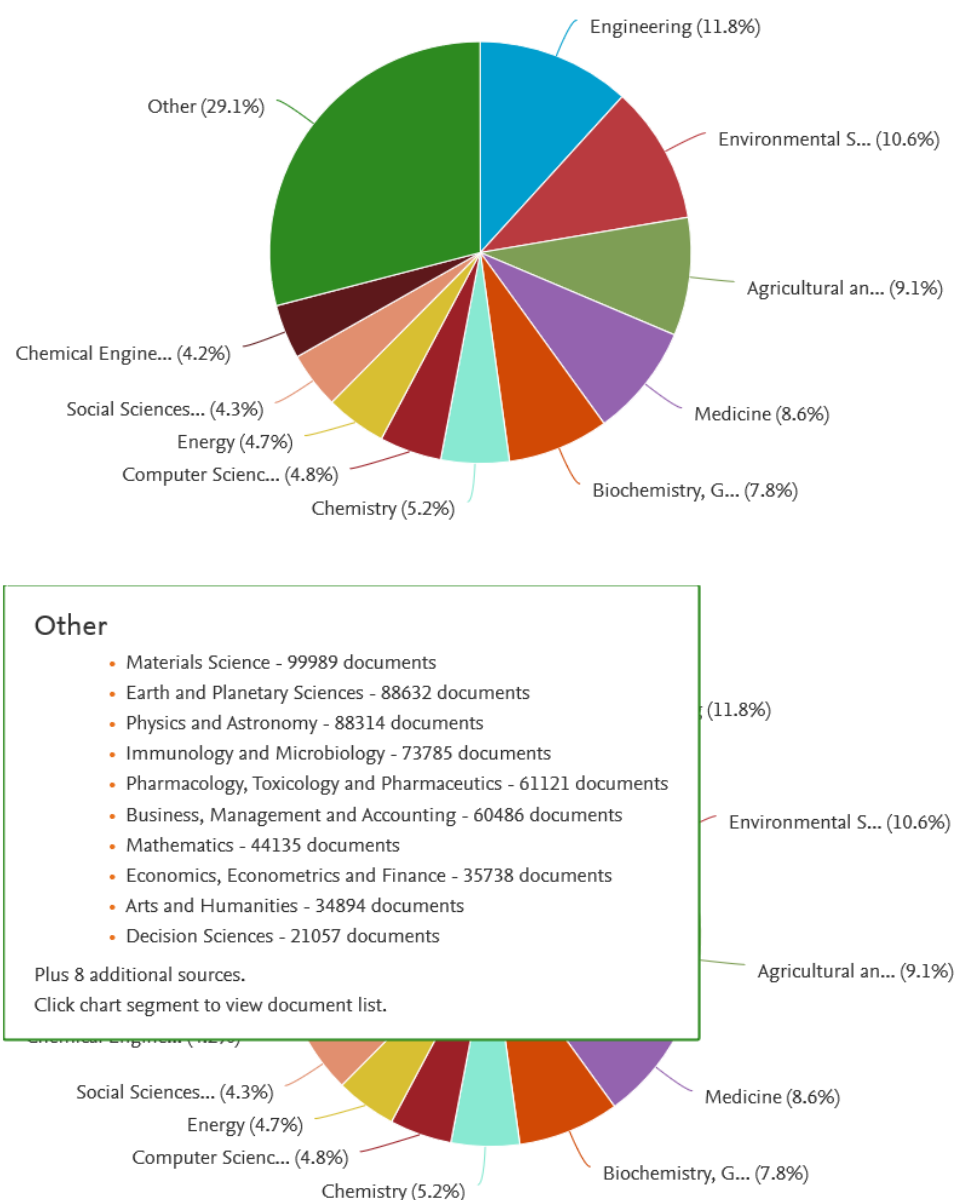
Title	Year	Source title	Cited by	EID
SHELXT - Integrated space-group and crystal-structure determination	2015	Acta Crystallographica Section A: Foundations of Crystallography	8529	s2.0-84920370689
Internet of Things (IoT): A vision, architectural elements, and future directions	2013	Future Generation Computer Systems	3941	2-s2.0-84876943063
The task force for the management of arterial hypertension of the european society of hypertension (esh) and of the european society of cardiology (esc)	2013	Journal of Hypertension	3222	2-s2.0-84880015206
Standards of medical care in diabetes-2014	2014	Diabetes Care	2881	2-s2.0-84892649479
High-resolution global maps of 21st-century forest cover change	2013	Science	2832	2--s2.0-84887724460
Dye-sensitized solar cells with 13% efficiency achieved through the molecular engineering of porphyrin sensitizers	2014	Nature Chemistry	2715	s2.0-84895119853
Massive MIMO for next generation wireless systems	2014	IEEE Communications Magazine	2694	2-s2.0-84896823919
Carbon nanotubes: Present and future commercial applications	2013	Science	2591	2-s2.0-84873808704
Diet rapidly and reproducibly alters the human gut microbiome	2014	Nature	2414	2-s2.0-84892828465
A polymer tandem solar cell with 10.6% power conversion efficiency	2013	Nature Communications	2348	2-s2.0-84874611797
Ultrasensitive photodetectors based on monolayer MoS ₂	2013	Nature Nanotechnology	2191	2-s2.0-84880234625
Bounding the role of black carbon in the climate system: A scientific assessment	2013	Journal of Geophysical Research Atmospheres	1996	2-s2.0-84875359259
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1794	2-s2.0-84924528297

Phyloseq: An R Package for Reproducible Interactive Analysis and Graphics of Microbiome Census Data	2013	PLoS ONE	1752	2-s2.0-84876427223
Guidelines for the prevention of stroke in patients with stroke and transient ischemic attack: A guideline for healthcare professionals from the American Heart Association/American Stroke Association	2014	Stroke	1703	2-s2.0-84903782410
Recent advances in semiconductors for photocatalytic and photoelectrochemical water splitting	2014	Chemical Society Reviews	1666	2-s2.0-84902144692
Noble metal-free hydrogen evolution catalysts for water splitting	2015	Chemical Society Reviews	1631	2-s2.0-84934916291
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1624	2-s2.0-84922433390
Rechargeable lithium-sulfur batteries	2014	Chemical Reviews	1608	2-s2.0-84912542845
Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: A microbiological and molecular biological study	2016	The Lancet Infectious Diseases	1489	2-s2.0-84957433508

It was not possible to draw a randomised sample from the search results. The high number of search results led to a time out in the SDG analytics tool. However, the search query consists of a variety of broad keywords and the search results in Scopus showed that many articles were included that are not closely related to the theme of the SDG.

General analysis - reflection on subject areas that should be excluded (incl. pie charts)

Documents by subject area



The general analysis shows the high variety of topics and articles that are included in the search results. The theme of the SDG has the potential to cover several fields, e.g. 'Engineering', 'Environmental Sciences' or 'Chemistry'. But the fields of 'Medicine', 'Biochemistry, Genetics and Molecular Biology' and 'Computer Sciences' also provide a significant number of articles. These fields are expected to provide articles that are not closely related to the theme of the SDG. This is confirmed by the findings of the analysis of the top 20 cited articles. This search query could be improved to retrieve better and more relevant articles. A first step would be to clean up the original query by narrowing down keywords and by deleting keywords that retrieve articles that are not directly relevant for this SDG.

Update 1) The search keywords were edited to remove the broader search terms, or to make them more specific by combining with other search keywords.

Original search string: 1,367,789 document results

TITLE-ABS-KEY(({environmental justice} OR {environmental damage} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR pollution OR pollutant* OR ozone OR pesticide* OR {life cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {natural resource} OR {natural resources} OR {resource footprint} OR {ecological footprint} OR {solid waste} OR {transportation cost} OR {transportation costs} OR {transport cost} OR {transport costs} OR post-harvest OR {waste generation} OR {corporate report} OR {corporate reports} OR {corporate sustainability} OR consumption OR production OR {business practice} OR {business practices} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR deforest* OR recycl* OR touris* OR {natural product} OR {natural products} OR {biobased product} OR {biobased products} OR {bio-based product} OR {bio-based products} OR {supply chain} OR {zero waste})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Rationale for edits to the search keywords:

<u>query term</u>	<u>decision</u>	<u>rationale</u>
{environmental justice]	Deleted	Not directly related to sustainable consumption or production, more on equality which is covered by SDG 16
Pollution	Changed to {nutrient pollution} OR {chemical pollution}	Pollution was too broad and led to a variety of hits in irrelevant fields, narrowed down.
Pollutant*	Deleted	Too broad, included in nutrient pollution and chemical pollution
Natural resource	Changed to {natural resource management}	Too broad as a keyword, will result in all types of hits, changed to natural resource management
Natural resources	Deleted	Included in natural resource management
Resource footprint	Changed to {environmental footprint}	Changed to a more commonly used keyword
Transportation costs (and variants)	Deleted	Included as it is an element of a life cycle assessment / analysis; will lead to irrelevant hits as a standalone query term
Transportation cost	Deleted	Included as it is an element of a life cycle assessment / analysis; will lead to irrelevant hits as a standalone query term
Transport cost	Deleted	Included as it is an element of a life cycle assessment / analysis; will lead to irrelevant hits as a standalone query term
Post-harvest	Changed to {harvest efficiency}	Post-harvest is too broad, narrowed down
Corporate report	Changed to {corporate social responsibility}	The keyword 'corporate report' on its own is too broad, changed to 'corporate social responsibility' as auditing and reports are part of CSR and CSR is a common theme within sustainable production / management
Corporate reports	Deleted	The keyword 'corporate reports' on its own is too broad, part of 'corporate social responsibility' as auditing and reports are part of CSR and CSR is a common theme within sustainable production / management
Consumption	ADD: (consumption AND (sustainable or spill or efficiency or behaviour or resource or resources)	Narrowed down
Production	ADD: (production AND (sustainable or spill or efficiency or behaviour or resource or resources)	Narrowed down

Business practice	Deleted	
Business practices	Deleted	
Touris*	Deleted	Not directly related to the SDG theme
{Life cycle assessment}	Added	Added as a variation on life cycle analysis
{wireless sensor network}	Added	AND NOT {wireless sensor network} to exclude IT
{wireless sensor networks}	Added	AND NOT {wireless sensor networks} to exclude IT
{wireless network}	Added	AND NOT {wireless network} to exclude IT
{wireless networks}	Added	AND NOT {wireless networks} to exclude IT
{astrophysics}	Added	AND NOT {astrophysics} to exclude results on space / star recycling

Updated search string (first update): 122,229 document results

TITLE-ABS-KEY ({environmental damage} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR {nutrient pollution} OR {chemical pollution} OR ozone OR pesticide* OR {life cycle assessment} OR {life cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {natural resource management} OR {environmental footprint} OR {ecological footprint} OR {harvest efficiency} OR {solid waste} OR {waste generation} OR {corporate social responsibility} OR {corporate sustainability} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR deforest* OR recycl* OR {natural product} OR {natural products} OR {biobased product} OR {biobased products} OR {bio-based product} OR {bio-based products} OR {supply chain} OR {zero waste} OR consumption AND (sustainable OR spill OR efficiency OR resource OR resources) OR pollution AND (sustainable OR spill OR efficiency OR resource OR resources) AND NOT ({wireless sensor network} OR {wireless sensor networks} OR {wireless network} OR {wireless networks} OR {astrophysics})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 122,229 document results, compared to 1,367,789 document results for the original search.

The top 20 cited articles were reviewed. Twelve out of the top 20 cited articles do not closely reflect the theme of the SDG. Specifically, these relate to IT (despite excluding articles on WSNs), astrophysics, medicine and innovation.

Top 20 most cited articles listed in Scopus from search string v1:

***red text shows articles which do not closely match the theme of the SDG**

Title	Year	Source title	Cited by	EID
Massive MIMO for next generation wireless systems	2014	IEEE Communications Magazine	2694	2-s2.0-84896823919
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1794	2-s2.0-84924528297
Spatially sparse precoding in millimeter wave MIMO systems	2014	IEEE Transactions on Wireless Communications	1002	2-s2.0-84897915272

Fullerene-Free Polymer Solar Cells with over 11% Efficiency and Excellent Thermal Stability	2016	Advanced Materials	969	2-s2.0-84971316579
Enzyme immobilisation in biocatalysis: Why, what and how	2013	Chemical Society Reviews	913	2-s2.0-84880122011
The IUPHAR/BPS Guide to PHARMACOLOGY in 2016: Towards curated quantitative interactions between 1300 protein targets and 6000 ligands	2016	Nucleic Acids Research	909	2-s2.0-84976865494
Applications of nanotechnology in water and wastewater treatment	2013	Water Research	810	2-s2.0-84879158907
Recycling of rare earths: A critical review	2013	Journal of Cleaner Production	803	2-s2.0-84879926400
InP nanowire array solar cells achieving 13.8% efficiency by exceeding the ray optics limit	2013	Science	775	2-s2.0-84874355840
An overview of current status of carbon dioxide capture and storage technologies	2014	Renewable and Sustainable Energy Reviews	728	2-s2.0-84905389111
Organic and inorganic contaminants removal from water with biochar, a renewable, low cost and sustainable adsorbent - A critical review	2014	Bioresource Technology	728	2-s2.0-84899994793
Introducing the illustris project: Simulating the coevolution of dark and visible matter in the universe	2014	Monthly Notices of the Royal Astronomical Society	692	s2.0-84913618569
Bee declines driven by combined Stress from parasites, pesticides, and lack of flowers	2015	Science	690	2-s2.0-84927566064
Decision aids for people facing health treatment or screening decisions	2014	Cochrane Database of Systematic Reviews	686	2-s2.0-84983646532
Nano-magnetite (Fe ₃ O ₄) as a support for recyclable catalysts in the development of sustainable methodologies	2013	Chemical Society Reviews	685	s2.0-84876950303
A survey of energy-efficient wireless communications	2013	IEEE Communications Surveys and Tutorials	640	2-s2.0-84873739593
Catalytic C-C bond-forming multi-component cascade or domino reactions: Pushing the boundaries of complexity in asymmetric organocatalysis	2014	Chemical Reviews	603	2-s2.0-84897671012
Biomimetic super-lyophobic and super-lyophilic materials applied for oil/water separation: A new strategy beyond nature	2015	Chemical Society Reviews	602	2-s2.0-84916939235
Nanotechnology in the real world: Redeveloping the nanomaterial consumer products inventory	2015	Beilstein Journal of Nanotechnology	595	2-s2.0-84945955942
A literature and practice review to develop sustainable business model archetypes	2014	Journal of Cleaner Production	580	2-s2.0-84893788213

In addition, a random sample of 20 articles has been analysed. Twelve of the randomised sample of 20 articles do not closely match the theme of the SDG. These were mainly related to energy sciences, IT and healthcare.

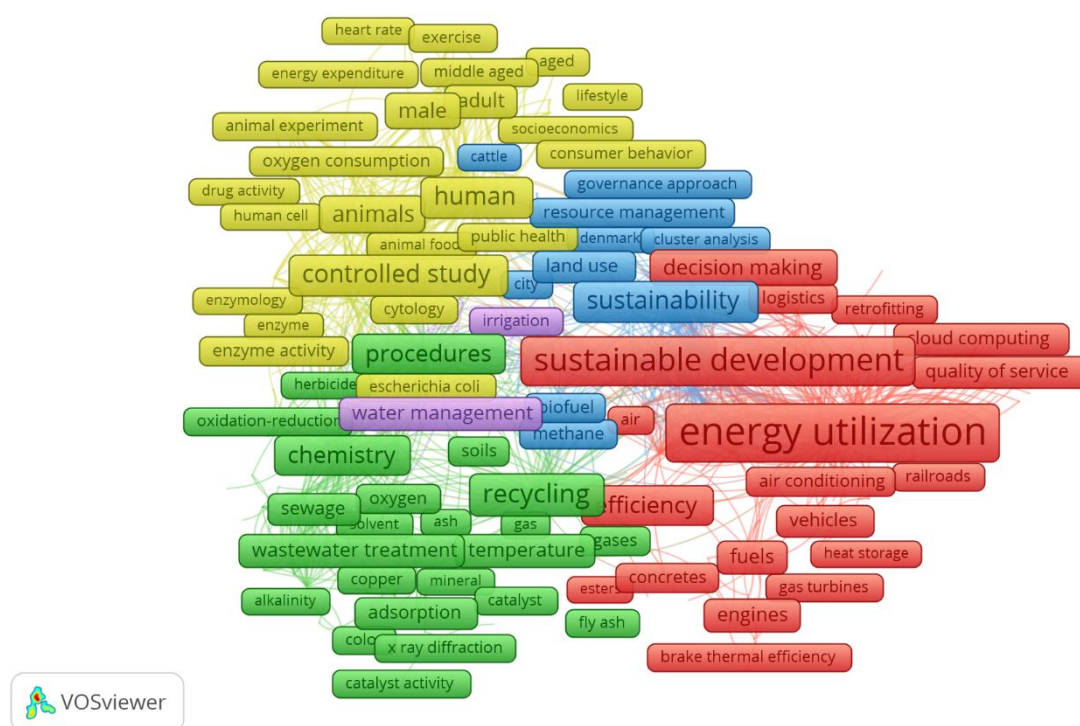
Randomised sample of 20 articles listed in Scopus from search string v1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Organizational Update	2015	Stroke	142	2-s2.0-84937217413
Efficacy of microalgae for industrial wastewater treatment: a review on operating conditions, treatment efficiency and biomass productivity	2016	Reviews in Environmental Science and Biotechnology	25	2-s2.0-84966340033
Concepts and methodologies for measuring the sustainability of cities	2014	Annual Review of Environment and Resources	18	2-s2.0-84908037398
Social-ecological predictors of global invasions and extinctions	2013	Ecology and Society	7	2-s2.0-84884957026
Dietary fatty acids influence sperm quality and function	2017	Andrology	6	2-s2.0-85017188221
A Model for Migration of Virtual Machines to Reduce Electric Energy Consumption	2016	Proceedings - 2016 10th International Conference on Complex, Intelligent, and Software Intensive Systems, CISIS 2016	6	2-s2.0-85011036664
A green synthesis of Pd nanoparticles supported on modified montmorillonite using aqueous: Ocimum sanctum leaf extract: A sustainable catalyst for hydrodechlorination of 4-chlorophenol	2016	RSC Advances	6	2-s2.0-84997545139
Ideals and institutions: Systemic reasons for the failure of a social forestry program in south-west Bangladesh	2016	Geoforum	4	2-s2.0-84995469444
An induction hardening process model to assist sustainability assessment of a steel bevel gear	2015	International Journal of Advanced Manufacturing Technology	4	2-s2.0-84949983199
Horizontal and vertical resource dilemmas in natural resource management: The case of African fisheries	2013	Fish and Fisheries	4	2-s2.0-84886376322
What Drives Managerial Perks? An Empirical Test of Competing Theoretical Perspectives	2015	Journal of Business Ethics	3	2-s2.0-84949317531
Evaluating the Energy Efficiency of Data Management Systems	2015	Proceedings - 4th International Workshop on Green and Sustainable Software, GREENS 2015	3	2-s2.0-84954471172
REDD+ initiatives for safeguarding biodiversity and ecosystem services: harmonizing sets of standards for national application	2014	Journal of Forest Research	3	2-s2.0-84910114692

ANTS: Pushing the rapid event notification in wireless sensor and actor networks	2013	2013 International Joint Conference on Awareness Science and Technology and Ubi-Media Computing: Can We Realize Awareness via Ubi-Media?, iCAST 2013 and UMEDIA 2013	2	2-s2.0-84899448223
Reducing energy consumption by using smart metering intelligent systems	2016	Cybernetics and Information Technologies	1	2-s2.0-84978895998
Limiting metabolic steps in the utilization of D-xylose by recombinant <i>Ralstonia eutropha</i> W50-EAB	2015	Wei sheng wu xue bao = Acta microbiologica Sinica	1	s2.0-84931086153
Development of energy-aware mobile applications based on resource outsourcing	2014	International Journal of Software Engineering and Knowledge Engineering	1	2-s2.0-84929325219
Investigating building sustainability by applying sensitivity analysis of impact factors during design stage	2017	Congress on Computing in Civil Engineering, Proceedings	0	2-s2.0-85021765638
Sequential treatments of oil sand tailings from green extraction by settling, ozonation, and sand filtration	2015	Sustainable Environment Research	0	s2.0-84929176810
New type of DLP optical engine equipped with the color LED and the compound parabolic concentrator	2013	International Journal of Innovative Computing, Information and Control	0	2-s2.0-84886391914

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the original search results to check if the updated search was primarily removing irrelevant results.

The original search query and the updated query were combined using AND NOT function in Scopus. This returned 1,245,560 document results – i.e. the publications from the first search that were excluded in the second search.

The top 20 most cited articles were reviewed. 19 of the 20 articles were not closely related to the theme of the SDG.

Title	Year	Source title	Cited by	EID
SHELXT - Integrated space-group and crystal-structure determination	2015	Acta Crystallographica Section A: Foundations of Crystallography	8529	2-s2.0-84920370689
Internet of Things (IoT): A vision, architectural elements, and future directions	2013	Future Generation Computer Systems	3941	2-s2.0-84876943063
The task force for the management of arterial hypertension of the european society of hypertension (esh) and of the european society of cardiology (esc)	2013	Journal of Hypertension	3222	2-s2.0-84880015206
Standards of medical care in diabetes-2014	2014	Diabetes Care	2881	2-s2.0-84892649479
High-resolution global maps of 21st-century forest cover change	2013	Science	2832	2-s2.0-84887724460
Dye-sensitized solar cells with 13% efficiency achieved through the molecular engineering of porphyrin sensitizers	2014	Nature Chemistry	2715	2-s2.0-84895119853
Carbon nanotubes: Present and future commercial applications	2013	Science	2591	2-s2.0-84873808704

Diet rapidly and reproducibly alters the human gut microbiome	2014	Nature	2414	2-s2.0-84892828465
A polymer tandem solar cell with 10.6% power conversion efficiency	2013	Nature Communications	2348	2-s2.0-84874611797
Ultrasensitive photodetectors based on monolayer MoS ₂	2013	Nature Nanotechnology	2191	2-s2.0-84880234625
Bounding the role of black carbon in the climate system: A scientific assessment	2013	Journal of Geophysical Research Atmospheres	1996	2-s2.0-84875359259
Phyloseq: An R Package for Reproducible Interactive Analysis and Graphics of Microbiome Census Data	2013	PLoS ONE	1752	2-s2.0-84876427223
Guidelines for the prevention of stroke in patients with stroke and transient ischemic attack: A guideline for healthcare professionals from the American Heart Association/American Stroke Association	2014	Stroke	1703	2-s2.0-84903782410
Recent advances in semiconductors for photocatalytic and photoelectrochemical water splitting	2014	Chemical Society Reviews	1666	2-s2.0-84902144692
Noble metal-free hydrogen evolution catalysts for water splitting	2015	Chemical Society Reviews	1631	2-s2.0-84934916291
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1624	2-s2.0-84922433390
Rechargeable lithium-sulfur batteries	2014	Chemical Reviews	1608	2-s2.0-84912542845
Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: A microbiological and molecular biological study	2016	The Lancet Infectious Diseases	1489	2-s2.0-84957433508
Natural Products as Sources of New Drugs from 1981 to 2014	2016	Journal of Natural Products	1474	2-s2.0-84962367258
High secondary aerosol contribution to particulate pollution during haze events in China	2015	Nature	1424	2-s2.0-84908317345

The update has removed a significant number of documents from the document results. The analysis of the document results that are excluded from the updated search results indicates that a large number of articles are rightfully excluded. The other analyses also show that another update is needed to improve the accuracy of the search query and to increase the relevance of the document results retrieved. Articles are still retrieved in the field of IT and in the field of medicine. Additionally, as the co-occurrence of the keywords indicates that articles in relation to the keyword 'energy utilization' forms an important element of the document results. However, this is not a specific target of the SDG. Improvements are needed to include articles that relate to (sustainable) resource utilization and other targets of the SDG. The following two updates are used to address these two issues: the exclusion of irrelevant fields (update 2) and an increased focus on the targets of the SDG while reducing the focus on energy utilization / efficiency (update 3).

Update 2) A second update was made to exclude articles from the field of medicine and IT.

The following keywords were added to the AND NOT query: {wireless}, {disease}

Search query version 2: 116,898 document results

TITLE-ABS-KEY ({environmental damage} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR {nutrient pollution} OR {chemical pollution} OR ozone OR pesticide* OR {life cycle assessment} OR {life

cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {natural resource management} OR {environmental footprint} OR {ecological footprint} OR {harvest efficiency} OR {solid waste} OR {waste generation} OR {corporate social responsibility} OR {corporate sustainability} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR deforest* OR recycl* OR {natural product} OR {natural products} OR {biobased product} OR {biobased products} OR {bio-based product} OR {bio-based products} OR {supply chain} OR {zero waste} OR consumption AND (sustainable OR spill OR efficiency OR resource OR resources) OR pollution AND (sustainable OR spill OR efficiency OR resource OR resources) AND NOT ({wireless sensor network} OR {wireless sensor networks} OR {wireless network} OR {wireless networks} OR {wireless} OR {disease} OR {astrophysics})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The top 20 cited articles were reviewed. Nine out of 20 articles do not closely match the theme of the SDG. These cover topics in the field of energy sciences, astrophysics and general engineering / production methods.

Top 20 most cited articles listed in Scopus from search string v2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1794	2-s2.0-84924528297
Fullerene-Free Polymer Solar Cells with over 11% Efficiency and Excellent Thermal Stability	2016	Advanced Materials	969	2-s2.0-84971316579
Enzyme immobilisation in biocatalysis: Why, what and how	2013	Chemical Society Reviews	913	2-s2.0-84880122011
Applications of nanotechnology in water and wastewater treatment	2013	Water Research	810	2-s2.0-84879158907
Recycling of rare earths: A critical review	2013	Journal of Cleaner Production	803	2-s2.0-84879926400
InP nanowire array solar cells achieving 13.8% efficiency by exceeding the ray optics limit	2013	Science	775	2-s2.0-84874355840
An overview of current status of carbon dioxide capture and storage technologies	2014	Renewable and Sustainable Energy Reviews	728	2-s2.0-84905389111
Organic and inorganic contaminants removal from water with biochar, a renewable, low cost and sustainable adsorbent - A critical review	2014	Bioresource Technology	728	2-s2.0-84899994793
Introducing the illustris project: Simulating the coevolution of dark and visible matter in the universe	2014	Monthly Notices of the Royal Astronomical Society	692	2-s2.0-84913618569
Bee declines driven by combined Stress from parasites, pesticides, and lack of flowers	2015	Science	690	2-s2.0-84927566064
Nano-magnetite (Fe ₃ O ₄) as a support for recyclable catalysts in the development of sustainable methodologies	2013	Chemical Society Reviews	685	2-s2.0-84876950303

Catalytic C-C bond-forming multi-component cascade or domino reactions: Pushing the boundaries of complexity in asymmetric organocatalysis	2014	Chemical Reviews	603	2-s2.0-84897671012
Biomimetic super-lyophobic and super-lyophilic materials applied for oil/water separation: A new strategy beyond nature	2015	Chemical Society Reviews	602	2-s2.0-84916939235
Nanotechnology in the real world: Redeveloping the nanomaterial consumer products inventory	2015	Beilstein Journal of Nanotechnology	595	2-s2.0-84945955942
A literature and practice review to develop sustainable business model archetypes	2014	Journal of Cleaner Production	580	2-s2.0-84893788213
Advances in Synthetic Applications of Hypervalent Iodine Compounds	2016	Chemical Reviews	547	2-s2.0-84960405249
A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems	2016	Journal of Cleaner Production	540	2-s2.0-84945954752
Recent developments in transition metal carbides and nitrides as hydrogen evolution electrocatalysts	2013	Chemical Communications	539	2-s2.0-84883854631
An overview of the environmental risks posed by neonicotinoid insecticides	2013	Journal of Applied Ecology	524	2-s2.0-84880326483
Functionalization of graphene for efficient energy conversion and storage	2013	Accounts of Chemical Research	524	2-s2.0-84872717597

In addition, a random sample of 20 articles was analysed. Eight of the randomised sample of 20 articles do not closely reflect the theme of the SDG. Specifically, these covered topics in the field of energy sciences (hit on efficiency), IT (hit on efficiency) and to archaeology (hit on resource).

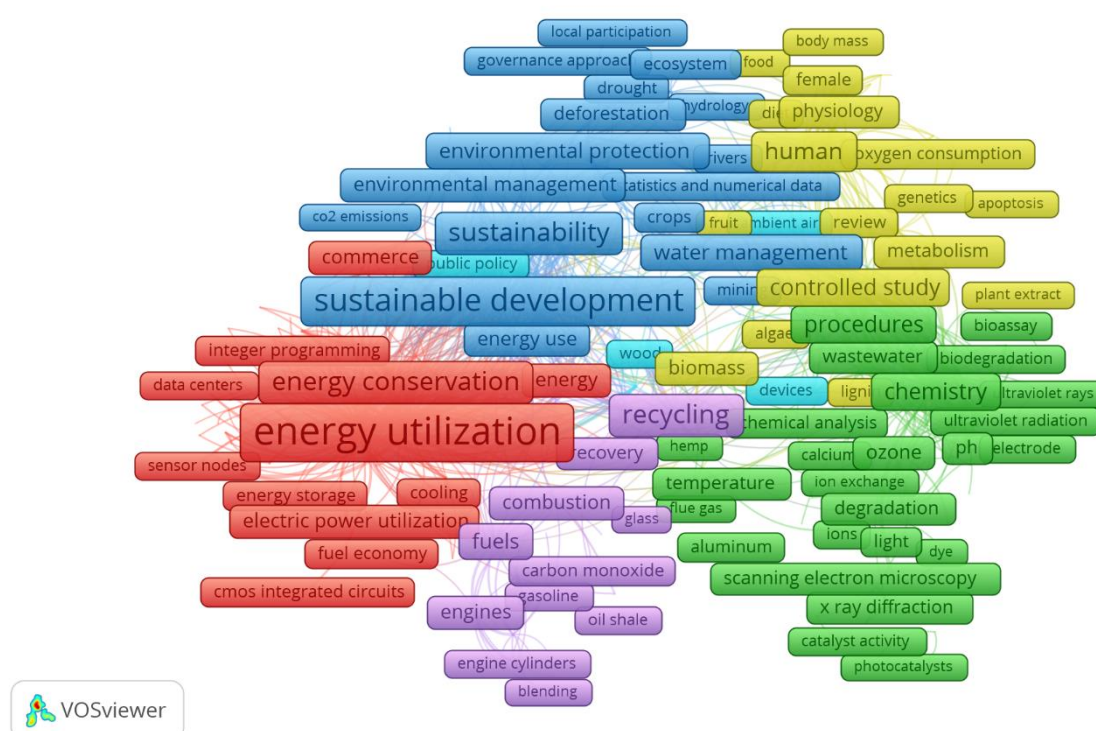
Randomised sample of 20 articles listed in Scopus from search string v2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Durable and scalable icephobic surfaces: Similarities and distinctions from superhydrophobic surfaces	2016	Soft Matter	85	2-s2.0-84957990322
Zero carbon energy system of South East Europe in 2050	2016	Applied Energy	55	2-s2.0-84962534495
Carbon Dioxide Utilisation: Closing the Carbon Cycle: First Edition	2014	Carbon Dioxide Utilisation: Closing the Carbon Cycle: First Edition	12	s2.0-84943571258
Photocatalytic degradation of phenol using Ag core-TiO ₂ shell (Ag@TiO ₂) nanoparticles under UV light irradiation	2016	Environmental Science and Pollution Research	10	2-s2.0-84946882044
Hydrometallurgical Processes for the Recovery of Metals from WEEE	2016	WEEE Recycling: Research, Development, and Policies	9	2-s2.0-85016988843

CO ₂ Emission Reduction Synergies of Advanced Engine Design and Fuel Octane Number	2014	SAE Technical Papers	8	2-s2.0-84938542438
Modeling of efficiency UHF class-D power amplifier with bandpass sigma-delta modulation	2017	2017 Systems of Signal Synchronization, Generating and Processing in Telecommunications, SINKHROINFO 2017	6	2-s2.0-85028574214
Revisiting crisis, change and institutions in the tropical forests: The multifunctional transition in Australia's wet tropics	2014	Journal of Rural Studies	5	2-s2.0-84904282070
The corporation: A critical, multi-disciplinary handbook	2017	The Corporation: A Critical, Multi-Disciplinary Handbook	3	2-s2.0-85048390841
Birds from the water: Reconstructing avian resource use and contribution to diet in prehistoric Scottish Island environments	2016	Journal of Archaeological Science: Reports	3	2-s2.0-84950263227
Emerging trends in groundwater pollution and quality	2014	Water Environment Research	3	2-s2.0-84937459647
Necessity of Including Maintenance Traffic Delay Emissions in Life Cycle Assessment of Pavements	2016	Procedia Engineering	1	2-s2.0-84999670070
Econology perspectives in naval industry	2013	International Multidisciplinary Scientific GeoConference Surveying Geology and Mining Ecology Management, SGEM	1	2-s2.0-84892591050
Oxidation efficiency of elemental mercury in two DBD plasma reactors	2013	Journal of Physics: Conference Series	1	2-s2.0-84875945071
A study of ratio and size H-zeolite catalyst for biolubricant oil production through three stages processes: Transesterification, epoxidation and ring opening modification	2017	Advanced Science Letters	0	2-s2.0-85027848488
Sustainable weed management using allelopathic approach	2017	Malaysian Applied Biology	0	2-s2.0-85021700233
Carbon capture and geological storage support policies in France [Politiques de soutien à la capture et au stockage géologique du carbone en France]	2015	Revue Economique	0	2-s2.0-84929083845
The heliostat design for the Natural Light Illumination System	2014	2014 International Conference on Advanced Robotics and Intelligent Systems, ARIS 2014	0	2-s2.0-84906700919
Balance routing protocol based on game theory for Ad Hoc networks	2014	Xibei Gongye Daxue Xuebao/Journal of Northwestern Polytechnical University	0	2-s2.0-84901240581
Research on application and development of the building life-cycle management in China	2013	Applied Mechanics and Materials	0	2-s2.0-84887203237

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the original search results to check if the updated search was primarily removing irrelevant results.

The search results retrieved by query version 1 and query version 2 were compared by using the AND NOT function in Scopus. The updated query led to the exclusion of 5,331 articles. 20 of the 20 top cited articles are not closely related to the theme of the SDG.

Title	Year	Source title	Cited by	EID
Massive MIMO for next generation wireless systems	2014	IEEE Communications Magazine	2694	2-s2.0-84896823919
Spatially sparse precoding in millimeter wave MIMO systems	2014	IEEE Transactions on Wireless Communications	1002	2-s2.0-84897915272
The IUPHAR/BPS Guide to PHARMACOLOGY in 2016: Towards curated quantitative interactions between 1300 protein targets and 6000 ligands	2016	Nucleic Acids Research	909	2-s2.0-84976865494
Decision aids for people facing health treatment or screening decisions	2014	Cochrane Database of Systematic Reviews	686	2-s2.0-84983646532
A survey of energy-efficient wireless communications	2013	IEEE Communications Surveys and Tutorials	640	2-s2.0-84873739593

Toward green and soft: A 5G perspective	2014	IEEE Communications Magazine	472	2-s2.0-84896808942
Google Earth Engine: Planetary-scale geospatial analysis for everyone	2017	Remote Sensing of Environment	454	2-s2.0-85021781951
Prevention of chronic disease in the 21st century: Elimination of the leading preventable causes of premature death and disability in the USA	2014	The Lancet	407	2-s2.0-84903778851
Safeguarding human health in the Anthropocene epoch: Report of the Rockefeller Foundation-Lancet Commission on planetary health	2015	The Lancet	360	2-s2.0-84947491070
A Survey on Mobile Edge Computing: The Communication Perspective	2017	IEEE Communications Surveys and Tutorials	349	2-s2.0-85028734877
Wireless information and power transfer: Energy efficiency optimization in OFDMA systems	2013	IEEE Transactions on Wireless Communications	334	2-s2.0-84891558256
CV protection in the EMPA-REG OUTCOME trial: A thrifty substrate hypothesis	2016	Diabetes Care	317	2-s2.0-84975853831
Efficiency of siRNA delivery by lipid nanoparticles is limited by endocytic recycling	2013	Nature Biotechnology	315	2-s2.0-84880274434
WHO's new end TB strategy	2015	The Lancet	307	2-s2.0-84929032442
Full duplex techniques for 5G networks: Self-interference cancellation, protocol design, and relay selection	2015	IEEE Communications Magazine	304	2-s2.0-84929303014
Myofibroblast-mediated mechanisms of pathological remodelling of the heart	2013	Nature Reviews Cardiology	283	2-s2.0-84871796377
Biology and management of Asian citrus psyllid, vector of the huanglongbing pathogens	2013	Annual Review of Entomology	244	2-s2.0-84873814207
Chloroquine in cancer therapy: A double-edged sword of autophagy	2013	Cancer Research	241	2-s2.0-84871940714
Energy-Efficient UAV Communication with Trajectory Optimization	2017	IEEE Transactions on Wireless Communications	226	2-s2.0-85020943053
Food choices, health and environment: Effects of cutting Europe's meat and dairy intake	2014	Global Environmental Change	224	2-s2.0-84901609714

Update 3) A third update was made to the search string to cover targets that are not (fully) covered yet.

The current query primarily focusses on pollution, waste treatment, a shift to non-carbon energy resources to reduce resource use and energy efficiency. Energy efficiency is not a specific target of this SDG, and hits on this topic should be reduced. The targets on material use, material efficiency and sustainable consumption require better coverage. Slight changes have been made to existing query terms by narrowing down keywords. The update was also used to add keywords on topics that are currently not covered yet.

Rationale for edits to the search keywords:

<u>query term</u>	<u>Decision</u>	<u>rationale</u>
consumption AND (sustainable OR spill OR efficiency OR resource OR resour ces)	Narrowed down to (consumption AND {resource use} OR spill))	
pollution AND (sustainable OR spill OR efficiency OR resource OR resources)	Narrowed down to (production AND {resource use} OR spill))	
Ozone	Narrowed down to {ozone depletion}	
Pesticide*	Narrowed down to {pesticide pollution} {pesticide stress} {pesticide reduction}	Narrowed down as the query term pesticide* primarily led to genetic and cell level studies / experiments. Keywords such as reduction / pollution / actual effects on the environment are more in line with the targets of the SDG
{environmental damage}	Changed to {environmental pollution}	Narrowed down to a target described in the SDG
{nutrient pollution}	Deleted	Not mentioned as a target, targets specifically mention chemicals. Deleted to reduce hits on irrelevant articles
{natural resource management}	Deleted	Deleted as it leads to too many hits on irrelevant articles and conservation management (ecology)
{ecological footprint}	Changed to {material footprint}	Changed to meet one of the key targets, ecological footprint is less useful as seen in the previous updates
Recycl*	Narrowed down to {waste recycling} OR {resource recycling} OR {resource reuse}	Narrowed down to limit hits on medical studies, astronomy, etc
Deforest*	Deleted	Deleted as it leads to too many hits on irrelevant articles and conservation management (ecology)
{natural product}	Deleted	A natural product or biobased product does not necessarily relate to sustainable resource use or a waste reduction. The query term also led to many hits on irrelevant articles. Deleted 5 variants, one variant was changed to {biobased economy} as the transition is seen as a step forward and progress is being made to make this type of economy more sustainable
{natural products}	Deleted	See {natural product}
{biobased product}	Deleted	See {natural product}
{biobased products}	Deleted	See {natural product}
{bio-based product}	Deleted	See {natural product}
{bio-based products}	Changed to {biobased economy}	See {natural product}

{supply chain}	Changed to {sustainable supply chain}	
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The table below gives an overview of the keywords that were added to enhance coverage of the search query.

Target	Keywords to be added
Implement the 10-year framework of programmes on sustainable consumption and production , all countries taking action, with developed countries taking the lead, taking into account the development and capabilities of developing countries	{sustainability label} {sustainability labelling}
By 2030, achieve the sustainable management and efficient use of natural resources	{global resource extraction} {material flow accounting} {societal metabolism}
By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses	{food spill} {resource spill} {resource efficiency} {sustainable food consumption} {green consumption}
By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle , in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment	Included
By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse	{circular economy} {cradle to cradle}
Encourage companies , especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle	Included
Promote public procurement practices that are sustainable, in accordance with national policies and priorities	{sustainable procurement}
By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature	Included
Support developing countries to strengthen their scientific and technological capacity to move towards more sustainable patterns of consumption and production	Included
Develop and implement tools to monitor sustainable development impacts for sustainable tourism that creates jobs and promotes local culture and products	{sustainable tourism}
Rationalize inefficient fossil-fuel subsidies that encourage wasteful consumption by removing market distortions, in accordance with national circumstances, including by restructuring taxation and phasing out those harmful subsidies , where they exist, to reflect their environmental impacts, taking fully into account the specific needs and conditions of developing countries and minimizing the possible adverse impacts on their	{fossil-fuel subsidies} {fossil-fuel expenditure}

development in a manner that protects the poor and the affected communities	
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Third update search: 68,509 document results

TITLE-ABS-KEY ({environmental pollution} OR {hazardous waste} OR {hazardous chemical} OR {hazardous chemicals} OR {toxic chemical} OR {toxic chemicals} OR {chemical pollution} OR {ozone depletion} OR {pesticide pollution} OR {pesticide stress} OR {pesticide reduction} OR {life cycle assessment} OR {life cycle analysis} OR {life cycle analyses} OR {life-cycle analysis} OR {life-cycle analyses} OR {low carbon economy} OR {low-carbon economy} OR {environmental footprint} OR {material footprint} OR {harvest efficiency} OR {solid waste} OR {waste generation} OR {corporate social responsibility} OR {corporate sustainability} OR {consumer behavior} OR {consumer behaviors} OR {consumer behaviour} OR {consumer behaviours} OR {waste recycling} OR {resource recycling} OR {resource reuse} OR {biobased economy} OR {zero waste} OR {sustainability label} OR {sustainability labelling} OR {global resource extraction} OR {material flow accounting} OR {societal metabolism} OR {food spill} OR {resource spill} OR {resource efficiency} OR {sustainable food consumption} OR {green consumption} OR {sustainable supply chain} OR {circular economy} OR {cradle to cradle} OR {sustainable procurement} OR {sustainable tourism} OR {fossil-fuel subsidies} OR {fossil-fuel expenditure} OR {sustainability label} OR {sustainability labelling} OR (consumption AND ({resource use} OR spill)) OR (production AND ({resource use} OR spill)) AND NOT ({wireless sensor network} OR {wireless sensor networks} OR {wireless network} OR {wireless networks} OR {wireless} OR {disease} OR {astrophysics})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search query retrieves 68,509 document results, compared to 116,898 document results retrieved by search query version 2.

The top 20 cited articles were reviewed. Four of the 20 articles are not closely related to the theme of the SDG. Two articles cover energy sciences, one covers carbon dioxide storage technologies and one covers consumer theories.

Top 20 most cited articles listed in Scopus from search string v3:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Towards greener and more sustainable batteries for electrical energy storage	2015	Nature Chemistry	1812	2-s2.0-84924528297
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1656	2-s2.0-84922433390
Rechargeable lithium-sulfur batteries	2014	Chemical Reviews	1624	2-s2.0-84912542845
Polymeric Photocatalysts Based on Graphitic Carbon Nitride	2015	Advanced Materials	1272	2-s2.0-85027957371
Plastic waste inputs from land into the ocean	2015	Science	1264	2-s2.0-84923095173
Biochar as a sorbent for contaminant management in soil and water: A review	2014	Chemosphere	1146	2-s2.0-84893809674
Phytoremediation of heavy metals-Concepts and applications	2013	Chemosphere	981	2-s2.0-84884376957
A review of soil heavy metal pollution from mines in China: Pollution and health risk assessment	2014	Science of the Total Environment	817	2-s2.0-84884585471

Recycling of rare earths: A critical review	2013	Journal of Cleaner Production	811	2-s2.0-84879926400
An overview of current status of carbon dioxide capture and storage technologies	2014	Renewable and Sustainable Energy Reviews	738	2-s2.0-84905389111
Electrochemical advanced oxidation processes: Today and tomorrow. A review	2014	Environmental Science and Pollution Research	651	2-s2.0-84903588978
A review of working fluid and expander selections for organic Rankine cycle	2013	Renewable and Sustainable Energy Reviews	616	2-s2.0-84876537293
Biomimetic super-lyophobic and super-lyophilic materials applied for oil/water separation: A new strategy beyond nature	2015	Chemical Society Reviews	611	2-s2.0-84916939235
Catalytic C-C bond-forming multi-component cascade or domino reactions: Pushing the boundaries of complexity in asymmetric organocatalysis	2014	Chemical Reviews	607	2-s2.0-84897671012
Nanotechnology in the real world: Redeveloping the nanomaterial consumer products inventory	2015	Beilstein Journal of Nanotechnology	603	2-s2.0-84945955942
A literature and practice review to develop sustainable business model archetypes	2014	Journal of Cleaner Production	594	2-s2.0-84893788213
A review on circular economy: The expected transition to a balanced interplay of environmental and economic systems	2016	Journal of Cleaner Production	554	2-s2.0-84945954752
Adsorptive removal of hazardous materials using metal-organic frameworks (MOFs): A review	2013	Journal of Hazardous Materials	554	2-s2.0-84872613894
Metal-organic frameworks as platforms for clean energy	2013	Energy and Environmental Science	547	2-s2.0-84878871031
Black titanium dioxide (TiO ₂) nanomaterials	2015	Chemical Society Reviews	537	2-s2.0-84925635305

In addition, a randomised sample of 20 articles was reviewed. Three of the 20 articles do not closely reflect the theme of the SDG. These articles cover topics related to gaming and healthcare.

Randomised sample of 20 articles listed in Scopus from the search string v3:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Greenhouse gas emissions of packaged fluid milk production in Tehran	2014	Journal of Cleaner Production	21	2-s2.0-84905102932
Understanding collaborative consumption: Test of a theoretical model	2017	Technological Forecasting and Social Change	18	2-s2.0-85014539180
Examining the impact of league entry fees on online fantasy sport participation and league consumption	2013	European Sport Management Quarterly	10	2-s2.0-84879622975
Integration of environmental aspects into R&D inter-organizational projects management: Application of a life cycle-based method to the development of innovative windows	2016	Journal of Cleaner Production	8	2-s2.0-85027925057

Simulation of Combustion of Municipal Solid Waste and Coal in an Industrial-Scale Circulating Fluidized Bed Boiler	2017	Energy and Fuels	6	2-s2.0-85039064005
Development of lightweight aggregates from stone cutting sludge, plastic wastes and sepiolite rejections for agricultural and environmental purposes	2017	Journal of Environmental Management	5	2-s2.0-85020042864
Production and characterization of smokeless bio-coal briquettes incorporating plastic waste materials	2017	Environmental Technology and Innovation	4	2-s2.0-85026552671
Some reflections on agrarian prospects	2016	Economic and Political Weekly	4	2-s2.0-84959558300
Socioeconomic Effects of Reducing Household Carbon Footprints Through Meat Consumption Taxes	2016	Journal of Food Products Marketing	4	2-s2.0-84959179267
Assembled products: The key to more effective competition and antitrust oversight in health care	2016	Cornell Law Review	4	2-s2.0-84963638659
Designing Public–Private Partnerships for Development	2015	Business and Society	4	2-s2.0-84927590607
Turning unexploited food waste into biomethane distributed through local filling stations networks	2016	European Biomass Conference and Exhibition Proceedings	1	2-s2.0-85019736633
Proposal for an operational performance indicator to evaluate sorting and composting plants for municipal solid waste	2015	International Journal of Environment and Waste Management	1	2-s2.0-84928780152
Enhanced heterogeneous wet hydrogen peroxide catalytic oxidation performance of fly ash-derived zeolite by CuO incorporation	2017	Scientia Iranica	0	2-s2.0-85029123987
Urbanisation and its effect on risk factors associated with childhood diarrhoea in Mbour, Senegal: A visualisation	2017	Geospatial Health	0	2-s2.0-85048464872
Adsorption of arsenate (HAsO ₄ ²⁻) by the clay fraction of soils of the Keller and Barton Peninsulas, King George Island, Maritime Antarctic	2017	Revista Ciencia Agronomica	0	2-s2.0-85009788894
A Hybrid Stochastic Approach for Improving Uncertainty Analysis in the Design and Development of a Wind Turbine	2016	Procedia CIRP	0	2-s2.0-85007021847
Retention and efficacy of ultra-low volume pesticide applications on <i>Culex quinquefasciatus</i> (Diptera: Culicidae)	2015	Environmental Science and Pollution Research	0	2-s2.0-84945448508
Ex-situ bioremediation of hydrocarbon contaminated soil - An example from Oman	2014	Society of Petroleum Engineers - SPE Middle East Health, Safety, Environment and Sustainable Development Conference and Exhibition, MEHSE 2014	0	2-s2.0-84926154824

Temporary systems after the earthquake in L'Aquila	2014	WIT Transactions on the Built Environment	0	2-s2.0-84903202251
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Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the search results retrieved by search query version 2 to check if the updated search was primarily removing irrelevant results.

The search results retrieved by query version 2 and query version 3 were compared by using AND NOT function in Scopus. This returned 175,092 document results – i.e. the publications from the first search that were excluded in the second search.

The top 20 cited articles were reviewed. Fourteen of the 20 articles are rightfully excluded from the updated search results. Six articles were excluded that could be relevant for the theme of the SDG, these covered topics related to waste water treatment, catalysts for sustainable production, insecticides and the sharing economy.

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Fullerene-Free Polymer Solar Cells with over 11% Efficiency and Excellent Thermal Stability	2016	Advanced Materials	983	2-s2.0-84971316579
Enzyme immobilisation in biocatalysis: Why, what and how	2013	Chemical Society Reviews	925	2-s2.0-84880122011

Applications of nanotechnology in water and wastewater treatment	2013	Water Research	827	2-s2.0-84879158907
InP nanowire array solar cells achieving 13.8% efficiency by exceeding the ray optics limit	2013	Science	777	2-s2.0-84874355840
Organic and inorganic contaminants removal from water with biochar, a renewable, low cost and sustainable adsorbent - A critical review	2014	Bioresource Technology	744	2-s2.0-84899994793
Bee declines driven by combined Stress from parasites, pesticides, and lack of flowers	2015	Science	712	2-s2.0-84927566064
Introducing the illustris project: Simulating the coevolution of dark and visible matter in the universe	2014	Monthly Notices of the Royal Astronomical Society	710	2-s2.0-84913618569
Nano-magnetite (Fe ₃ O ₄) as a support for recyclable catalysts in the development of sustainable methodologies	2013	Chemical Society Reviews	689	2-s2.0-84876950303
Advances in Synthetic Applications of Hypervalent Iodine Compounds	2016	Chemical Reviews	562	2-s2.0-84960405249
Recent developments in transition metal carbides and nitrides as hydrogen evolution electrocatalysts	2013	Chemical Communications	544	2-s2.0-84883854631
Global obesity: Trends, risk factors and policy implications	2013	Nature Reviews Endocrinology	532	2-s2.0-84871090162
An overview of the environmental risks posed by neonicotinoid insecticides	2013	Journal of Applied Ecology	530	2-s2.0-84880326483
Functionalization of graphene for efficient energy conversion and storage	2013	Accounts of Chemical Research	526	2-s2.0-84872717597
Additive manufacturing and its societal impact: A literature review	2013	International Journal of Advanced Manufacturing Technology	510	2-s2.0-84888636647
A review of energy sources and energy management system in electric vehicles	2013	Renewable and Sustainable Energy Reviews	505	2-s2.0-84871785266
Catalysis for CO ₂ conversion: A key technology for rapid introduction of renewable energy in the value chain of chemical industries	2013	Energy and Environmental Science	503	2-s2.0-84878833421
The sharing economy: Why people participate in collaborative consumption	2016	Journal of the Association for Information Science and Technology	493	2-s2.0-84983357869
Recent advances in solid sorbents for CO ₂ capture and new development trends	2014	Energy and Environmental Science	476	2-s2.0-84907984288
Ecology and Management of Forest Soils: Fourth Edition	2013	Ecology and Management of Forest Soils: Fourth Edition	473	2-s2.0-84891585138
Sustainability: Four billion people facing severe water scarcity	2016	Science Advances	462	2-s2.0-84961611666

The different analyses show that this update provides an important improvement by bringing back the number of irrelevant document results, especially regarding articles related to energy efficiency. The top cited 20 and the randomised sample of 20 articles show that a low number of irrelevant articles are retrieved. This is confirmed by the co-occurrence of keywords, which visualises a better focus on the targets of the SDG. Especially in relation to waste, recycling, sustainable production (among which LCAs), consumer behaviour and environmental pollution.